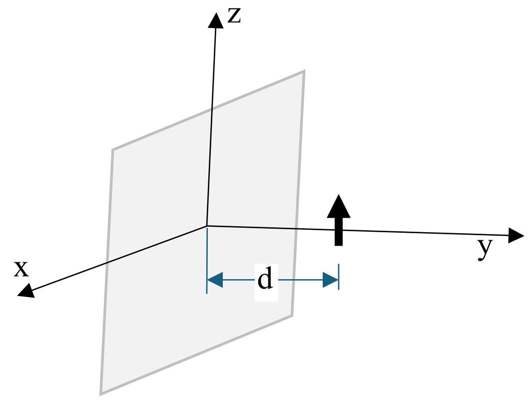
Read: 6.1, 6.2, 6.3, 6.4, 6.6, 6.7, 6.8.1, 6.8.2, 7.3, 7.4, 7.5, 7.8, 7.9

1. A z-directed current element *Il* on the y-axis a distance d from a ground plane in the x-z plane. Show the electric field in this configuration is:

$$E_\theta = \frac{-\eta Il}{\lambda}\frac{e^{-jkr}}{r}\sin\theta\sin(kd\sin\varphi\sin\theta)$$



2. In Notes 7 the total power radiated by a short vertical dipole over a perfectly conducting (PEC) ground was found to be (an extra 2 in the notes was incorrect – I neglected to put in the ½ in the initial time average power density term)

$$P_{rad} = \pi\eta\left(\frac{I\ell}{\lambda}\right)^2\left[\frac{1}{3}+\frac{\sin(2kd)}{(2kd)^3}-\frac{\cos(2kd)}{(2kd)^2}\right]$$

(a) Starting with the electric and magnetic fields show that the time average power density for the dipole over the ground can be written

$$P_{ave} = \frac{\eta}{2r^2}\left(\frac{I\ell}{\lambda}\right)^2\sin^2\theta\cos^2(kd\cos\theta)$$

(b) Show the directivity is $D_o = 2/T$ where $T = \left[\frac{1}{3}+\frac{\sin(2kd)}{(2kd)^3}-\frac{\cos(2kd)}{(2kd)^2}\right]$

(c) Show the radiation resistance is $R_r = 2\pi\eta\left(\frac{l}{\lambda}\right)^2 T$

3. Use the equivalence principle to find the integral equation for the current on a thin solid PEC cylinder of length *L* and radius *a* and illuminated by a plane wave. Neglect the current on the endcaps. Draw pictures to show how the equivalence principle is applied. Set up the complete problem including the integrals for the surface of the cylinder with the proper $|\mathbf{r}-\mathbf{r}'|$ terms, but do not evaluate them.